

Roadmap

PREPARED BY Energy Hippo implementation team

Phased plan to make Northlake the dream synthetic customer. Each phase lists the **deliverables** (what we add to the source model + generated artifacts) and the **acceptance criteria** (the EEMSuite reports/outputs that must render non-empty with a **known expected value**).

Phases express the eventual coverage backlog. Their prerequisites apply to the capability being exercised; the full user/permission matrix is not a blocker for the first gateway.

Current milestone — UI foundation and one gateway

Use the [packet 0.3 customer inventory](#) and [Student Center UI checklist](#):

1. With the existing setup account, create the missing Northlake company, site, building, provider, account, meter and two interval points through the UI.
2. Configure the AcquiSuite gateway and its normal workflow/task path through the UI. Capture actual instance IDs; scenario numeric IDs are illustrative.
3. Run the supplied complete-day, replay, gap and backfill cases. Confirm stored values and visible results, including 96 readings per point and 2,020 kWh.
4. Extend to the MDEF electric feed, aggregates, weather relationships and handheld register/usage pairs, then the remaining format cases.
5. Promote a database backup only after the applicable checks pass, retaining configuration, source fixtures, evidence and a successful restore check.

Source files and the customer packet are generated. UI entry and installed acceptance are still pending. Full rate modeling, bills, calendars, GL and user scenarios follow when their respective prerequisites are needed.

Phase 0 — Foundation specs (enabling)

Why first. Reports need hierarchy, appropriate access, units and meters ↔ accounts; billing also needs company calendars. This repo supplies the customer facts and UI-entry instructions. The first gateway uses the existing setup account; the complete permission exercise below is a later acceptance step. Review older permission specifications against the installed product before applying them.

Deliverables (specs / manifests)

- Source-model extensions ([implementation/synthetic-source-model.pdf](#)) for the substrate gaps: **bill charge lines, company calendar, concrete permission grants, digital/status points, notes/images**.
- A concrete permission-grant spec mapping the 16 semantic permission profiles to real EEMSuite Permissions_TBL types (\$/A/D/EX/F/MI/MU/TS) + SiEAppObjects + AccessLevel, including node-scoped D grants and an AccessLevel-3 approver vs. an AccessLevel-1 viewer.
- A 2024–2025 monthly CompanyCalendar spec per company, with 1–2 months deliberately omitted for the import-rejection scenario.

Acceptance (verified after UI implementation)

- Substrate reports render: `rp_BuildingList`, `rp_PointsList`, `rp_HierarchyDetailsEditor`, `rp_IndexList`, and the config screens.
- A non-admin user sees only their permitted nodes; an `AccessLevel-1` user cannot perform an `AccessLevel-3` action.

Phase 1 — Consumption & cost core

Goal. Light up everything that runs on bills + intervals + units — the bulk of day-to-day reporting.

Deliverables

- Extend the `bill` model with **charge-line items** (`BillItem`: commodity, quantity, unit, amount, GL placeholder) so bills decompose into energy/demand/ fixed/tax components, routed into the "Actual" `BillScenario`.
- Ensure ≥ 13 consecutive monthly bills per account; populate `StatusHistory` (entry dates, ≥ 2 distinct entry users) so late/daily-processing logic works.
- Load the existing interval/aggregate data; add a few **digital/status points** (chiller enable, occupancy) for `DigitalSummary`.
- Add notes + a couple of images on meters/bills for `NoteHistory/ImageManagement`.

Acceptance

- All 15 Use Analysis reports render with known totals (interval count, peak demand, EUI).
- Bill Processing/Tracking core renders: `AccountDetail/Summary/History`, `BillHistory`, `BillVariance(+Analysis/Detail/Ranking)`, `CostBreakOut`, `FacilityAccountDetail`, `FiscalPeriodBills`, `LateBills`, `DailyProcessingSummary`, `BillDataGaps`, `MonthlyUtilityDistribution`.
- Known expected rollups by customer/site/building/provider/account/meter.

Phase 2 — Rate engine completeness

Goal. Exercise all **9 determinant math kinds** and every rate construct.

Deliverables. Grow the rate model to determinants/blocks/schedules and author ~10 rate schedules, assigned to meters (`UtilityMeterRates`) over ≥ 13 months:

1. **Residential inclining-block** (Cedar Row) — tiered energy (200), fixed (400), minimum bill (500), state + municipal taxes (800).
2. **Commercial TOU + seasonal demand** (campus, flagship) — On/Off-peak energy, summer/winter demand, **holiday calendar**, a % rider (700) + per-unit rider (200), taxes (800).
3. **Large-demand ratchet** (Science Center / chiller) — ratchet demand + load factor; needs a strong summer peak so the winter floor binds.
4. **Base/excess contract demand** — two-part demand with pre/post **determinant scripts**.
5. **RTP / index** — type **100** referencing an hourly **price-signal point** (the only way to fire RTP), plus a price-range-capped variant.
6. **Stepped + max-bill** — types **210** and **600** (both unexercised anywhere).

7. **Gas with monthly fuel adjustment** — fixed-by-day (300), volumetric energy with **unit conversion**, monthly effective-dated fuel-adjustment blocks.
8. **Multi-commodity** water/sewer/CHW/steam — multiple measure types + unusual units (tons, klb) + `UtilityCommodityMap` for component mapping.
9. **Rate-class options** — voltage/phase/tax-exempt/farm selectable classes.
10. **Multi-currency** (optional) — a C\$/£ rate against a US\$ reporting profile.

Document **unsupported constructs** (power factor, coincident-peak/4-CP, CPP, net-metering banking) as explicit synthetic stand-ins.

Acceptance

- All 9 Rate Analysis reports render; `rp_RateComparison` shadow-bills; every `UtilityRates_Run_NNN` proc fires at least once; RTP/stepped/max-bill/ratchet each have a verifying scenario with a known computed total.

Phase 3 — AP / GL & financial interface

Goal. "Do actual billing" end to end and **produce real AP interface files.**

Deliverables

- A chart of GL expense/AP accounts per company + the 5 user-defined GL segments (`GLAccountUDField/Value`) with realistic + partly-blank coverage.
- GL-code the bill lines (`BillItem.BI_ExpAcctID`) and account defaults; set account `Type_Code` mix (1 AP / 2 GL / 3 both / 0 no-upload negative control).
- `ExportSetup` rows + builder procs to **generate one file of each of the 7 formats**: fixed-width AP (`export_ap`), CSV AP (`export_ap2`), GL-UDF header+detail (Sacramento County style), multi-record AP voucher + GL journal (PeopleSoft/Massport style), SAP FI CSV (Clark County style), AR interface.
- Bill validation thresholds + bills that pass and trip each rule family; ≥ 1 open alert that **blocks** AP export.
- Accruals: set the 2300/2301 commodities; one open accrual + one trued-up accrual $\leftrightarrow$ reversal pair.

Acceptance

- `rp_AccountsPayableDetail`, `rp_APApprovalAndUpload`, `rp_GLAccountUDV`, `my_exports` render.
- One example file of **each** AP/GL/AR format generated and traceable to bills.
- `rp_BillValidation` / `rp_BillValidationStatus` show passes + failures; `rp_OpenAccruals` shows the open accrual and its reversal.

Phase 4 — Tenant rebilling

Goal. Exercise **every** allocation + reconciliation path. Vehicle (owner decision): **both** a new mixed-use **Northlake Town Center** (commercial allocation) and sub-metered **Cedar Row** (residential allocation).

Deliverables

- A multi-tenant structure (anchor retail + in-line shops + office tower + food court + common area + a cost center) with the meters/sub-meters each method needs.

- A bill-allocation config per method: Fixed Amount & %, Fully Submetered, Partially Submetered, Building Areas, Coincident Demand, Metered Substations, and **Bill Aggregation** (children → cost center). Overhead in both combined and separate modes.
- WUI **point families**: Weighted Use Index + Measured Use, with deliberate fail-reason cases (no index, no area, partial data, broken chain, no rate); note the unimplemented "Percentage Based" method.
- Generated tenant bills under all 4 generation types; a reconciliation scenario (interval-source and bill-source) with a deliberate nonzero variance; tenant accruals.

Acceptance

- All 14 Tenant Rebilling reports + `rp_TenantMap` render.
- Each allocation method has a config that produces tenant bills summing back to the parent (penny-true-up); `rp_BillReconciliation` shows known over/under variance.

Phase 5 — Sustainability & analytics

Goal. GHG, ENERGY STAR ratings, weather regression, benchmarks.

Deliverables

- **GHG**: emission factor source (eGRID region + EPA stationary combustion), factor members with effective dates, meter → category mapping (electric Scope 2, gas/steam Scope 1), per-company scope config.
- **ENERGY STAR ratings**: obtain responses through the supported sandbox integration for the 13 mapped properties across several periods; retain request/response evidence.
- **Weather regression**: run the supported calculation workflow for metered buildings from interval + weather data and verify the resulting fit statistics.
- **Benchmarks**: EUI + cost-intensity benchmark definitions over the building peer set.

Acceptance

- GHG History/Variance/Audit render with computed emissions; ES Rating Comparison/History/RequestLog render with scores; `rp_weatherregression` / `rp_RegressionSummaryReport2` render fitted models; `rp_BenchmarkRanking` / `rp_MonthlyEUI` / `rp_PerformanceSnapshot` rank the peer set.

Phase 6 — Operations & engagement

Goal. The operational layer + scheduled delivery to **100+ users**.

Deliverables

- **Project tracking**: a portfolio (LED retrofit, chiller upgrade, solar expansion) with activities, financial events, and M&V linkage to baselines.
- **Alarms**: alarm-enabled points with raised/ack/resolved rows across the span.
- **Audit / activity**: enable audit; generate a user action + data-change trail.
- **Tasks / workflows**: schedules for gateways + bill import + delivery; a couple of multi-step workflows; TaskLog run history; GatewayStatus live feed.
- **MFR delivery**: a saved-report + dashboard library; delivery schedules whose group-based recipient lists resolve to **100+ users** (needs the scale decision), plus individual + external recipients; the delivery task.

- **EnergyAI:** pre-generated insight rows for a couple of load points.
- **HMR:** explicit meter/route/read-type config tied to the handheld uploads.
- **Importers:** bill-importer source + mappings + sample BIF/CSV/Excel (incl. the bad-data validation sample), a GreenButton ESPI XML feed, a MyUpload/MyImport template.

Acceptance

- Project, Alarm, Audit, ActivityTracker, TaskLog, GatewayStatus, EnergyAllInsights, HMRConfiguration, bill_importer, my_imports render.
- A scheduled MFR delivery fans out to 100+ recipients (verifiable via the delivery task log) across PDF/Excel.
- Every one of the 115 reports has at least one scenario that renders it non-empty.

Cross-cutting: assertions

Per the project quality bar, each phase also adds `expected_assertion` entries so downstream results are validated automatically (known totals, counts, variances, file checksums), keeping the dream customer a **regression** asset, not just a demo.